

Grade 10 Chemistry

Questions & Answers — New Syllabus

Based on: Science Part I Textbook (Educational Publications Department)

Q1

Topic: Structure of Matter

What are the three main subatomic particles of an atom? Describe their charges.

ANSWER

An atom is made up of three main subatomic particles:

1. Proton — found in the nucleus. It has a positive charge (+1).
2. Neutron — found in the nucleus. It has no charge (neutral).
3. Electron — revolves around the nucleus in energy levels (shells). It has a negative charge (-1).

In a neutral atom, the number of protons equals the number of electrons, so the overall charge is zero.

The nucleus is positively charged because it contains protons and neutrons.

Q2

Topic: Structure of Matter

Define atomic number and mass number. Find the number of neutrons in a sodium atom (atomic number = 11, mass number = 23).

ANSWER

Atomic Number (Z): The number of protons in the nucleus of an atom. It is unique to each element. In a neutral atom, the atomic number also equals the number of electrons.

Mass Number (A): The total number of protons and neutrons in the nucleus of an atom.

Solution — Finding neutrons in Sodium (Na):

Atomic number (Z) = 11, so protons = 11

Mass number (A) = 23

Number of neutrons = $A - Z = 23 - 11 = 12$

Therefore, a sodium atom has 12 neutrons.

Mass number (A) = Number of protons + Number of neutrons

Number of neutrons = $A - Z = 23 - 11 = 12$

Q3

Topic: Electronic Configuration

What is electronic configuration? Write the electronic configuration of Carbon (atomic number = 6) and Chlorine (atomic number = 17).

ANSWER

Electronic configuration is the arrangement that shows how electrons are distributed among the energy levels (shells) of an atom, starting from the shell nearest to the nucleus.

The maximum number of electrons in each shell:

Shell K (1st) = 2 electrons

Shell L (2nd) = 8 electrons

Shell M (3rd) = 18 electrons

Carbon (C) — Atomic number = 6:

6 electrons distributed as: 2 in K, 4 in L

Electronic configuration = 2, 4

Chlorine (Cl) — Atomic number = 17:

17 electrons distributed as: 2 in K, 8 in L, 7 in M

Electronic configuration = 2, 8, 7

Carbon (C, Z=6) : 2, 4

Chlorine (Cl, Z=17): 2, 8, 7

Q4

Topic: Modern Periodic Table

Explain how the Period and Group of an element are determined in the Modern Periodic Table. In which Period and Group is Magnesium (atomic number = 12)?

ANSWER

The Modern Periodic Table is based on atomic number and electronic configuration.

Period: The Period to which an element belongs is determined by the number of energy levels (shells) that contain electrons in its atom.

Group: The Group to which an element belongs is determined by the number of electrons in its outermost energy level (valence electrons).

Magnesium (Mg) — Atomic number = 12:

Electronic configuration = 2, 8, 2

Number of occupied energy levels = 3 → Period 3

Electrons in outermost level = 2 → Group II

Therefore, Magnesium is in Period 3, Group II of the Periodic Table.

Mg (Z=12): Electronic config = 2, 8, 2

Period = 3 | Group = II

Q5

Topic: Structure of Matter — Isotopes

What are isotopes? Give two examples and explain why isotopes of the same element have the same atomic number but different mass numbers.

ANSWER

Isotopes are atoms of the same element that have the same number of protons (same atomic number) but different numbers of neutrons (different mass numbers).

Since isotopes of the same element have the same number of protons, their atomic number is identical. However, because they differ in neutron count, their mass numbers differ.

Example 1 — Hydrogen has 3 isotopes:

Protium: 1 proton, 0 neutrons — mass number = 1

Deuterium: 1 proton, 1 neutron — mass number = 2

Tritium: 1 proton, 2 neutrons — mass number = 3

Example 2 — Chlorine has 2 isotopes:

Cl-35: 17 protons, 18 neutrons — mass number = 35 (75% abundance)

Cl-37: 17 protons, 20 neutrons — mass number = 37 (25% abundance)

Atomic number = same (protons unchanged)

Mass number = different (neutron count varies)

Q6

Topic: Chemical Bonds — Ionic Bonds

What is an ionic bond? Explain how an ionic bond is formed between sodium (Na) and chlorine (Cl) to form sodium chloride (NaCl).

ANSWER

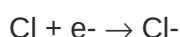
An ionic bond is the strong electrostatic attractive force formed between a positively charged ion (cation) and a negatively charged ion (anion) as a result of the transfer of electrons from one atom to another.

Formation of NaCl:

Step 1 — Sodium (Na) has electronic configuration 2, 8, 1. It has low electronegativity and tends to lose its one outer electron, forming a Na⁺ ion (cation):



Step 2 — Chlorine (Cl) has electronic configuration 2, 8, 7. It has high electronegativity and gains one electron to complete its outer shell, forming a Cl⁻ ion (anion):



Step 3 — The oppositely charged Na⁺ and Cl⁻ ions attract each other by electrostatic force, forming the ionic compound sodium chloride (NaCl). This is an ionic (electrovalent) bond.

In solid NaCl, millions of ions arrange in a regular 3D structure called an ionic lattice.

$\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$ (loses 1 electron)

$\text{Cl} + \text{e}^- \rightarrow \text{Cl}^-$ (gains 1 electron)

$\text{Na}^+ + \text{Cl}^- \rightarrow \text{NaCl}$ (ionic bond)

Q7

Topic: Chemical Bonds — Covalent Bonds

What is a covalent bond? How does it differ from an ionic bond? Give two examples of covalent compounds.

ANSWER

A covalent bond is formed when two atoms share a pair of electrons between them, so that each atom achieves a stable noble gas electronic configuration.

Covalent bonds form between atoms of similar or equal electronegativity (usually non-metals).

Key differences:

Ionic bond: formed by the TRANSFER of electrons from one atom to another, resulting in oppositely charged ions held together by electrostatic forces. Example: NaCl.

Covalent bond: formed by the SHARING of electrons between two atoms, with no ions formed. Example: H₂O, CH₄.

Two types of covalent molecules:

1. Homoatomic — sharing between atoms of the same element: H₂, F₂, O₂, N₂
2. Heteroatomic — sharing between atoms of different elements: H₂O (water), CH₄ (methane), NH₃ (ammonia)

Example — Fluorine molecule (F₂):

Each F atom has configuration 2, 7. By sharing one electron each, both atoms achieve the stable configuration 2, 8, forming a single covalent bond: F-F.

Ionic bond: electron TRANSFER → ions formed (e.g. NaCl)

Covalent bond: electron SHARING → no ions formed (e.g. H₂O, CH₄)